

# MACHINE LEARNING RESEARCH PROPOSAL

## Title

**E-XGBoost: A SHAP-Guided Feature-Pruned XGBoost Framework for Explainable Network Intrusion Detection**

## Keywords

1. Network Intrusion Detection
2. XGBoost
3. Explainable Artificial Intelligence (XAI)
4. SHAP
5. Cybersecurity
6. Feature Pruning

---

## 0.2 Identification Table

| Field            | Your Entry                                         |
|------------------|----------------------------------------------------|
| Researcher Name  | Frederick Jeremy Kwofie                            |
| Student ID       | 20970295                                           |
| Supervisor       | Dr. Eric                                           |
| Department       | Department of Computer Science                     |
| Institution      | Kwame Nkrumah University of Science and Technology |
| Submission Date  | 25/06/202                                          |
| Research Sector  | Cybersecurity                                      |
| Target Journal 1 | To Be Determined                                   |
| Target Journal 2 | To Be Determined                                   |

---

## NOVELTY DECLARATION

Machine learning-based Intrusion Detection Systems have demonstrated strong performance in detecting cyberattacks. However, many existing XGBoost-based approaches utilize all available network traffic features without systematically evaluating their contribution to classification performance. This often increases computational complexity and limits model interpretability.

This study proposes E-XGBoost, an engineered intrusion detection framework that removes low-contribution features using SHAP global importance thresholds before model training. The framework evaluates the trade-off between detection performance, inference efficiency, and interpretability using both CIC-IDS2017 and institutionally collected network traffic logs.

### **Novelty Type**

- ☑ New Model Combination / Hybrid Architecture
  - ☑ New Explainability Framework
  - ☑ New Evidence in Cybersecurity Applications
- 

## **SECTION 1 – SECTOR & TOPIC SELECTION**

### **Research Sector**

Cybersecurity

### **Specific Sub-Problem**

Accurate and explainable detection of malicious network traffic using machine learning techniques while reducing computational complexity through feature optimization.

---

## **SECTION 2 – PROBLEM STATEMENT, OBJECTIVES & THEORY**

### **2.1 Background / Context**

Cybersecurity threats continue to increase as organizations become increasingly dependent on digital infrastructure and networked systems. Intrusion Detection Systems (IDS) play a critical role in protecting information assets by identifying malicious activities and unauthorized access attempts. Traditional signature-based intrusion detection approaches are effective against known threats but often struggle to detect previously unseen attacks, creating a need for more intelligent detection mechanisms (Sharafaldin et al., 2018).

Machine learning has emerged as a promising solution because of its ability to learn patterns from network traffic data and identify anomalies that may indicate cyberattacks. Among the available machine learning algorithms, XGBoost has demonstrated strong predictive performance, scalability, and robustness across a wide range of classification tasks (Chen & Guestrin, 2016). Recent research has also shown that explainable artificial intelligence techniques can improve transparency and trust in intrusion detection systems by helping security analysts understand how machine learning models arrive at their decisions (Ahmed et al., 2024; Setiawan et al., 2026).

Despite these advances, many intrusion detection models still operate as black-box systems, making it difficult for analysts to interpret predictions and justify security decisions. Consequently, there is a growing need for intrusion detection frameworks that combine high predictive performance with explainability and computational efficiency.

### **2.1.1 Theoretical Framework**

This study is guided by the Defense-in-Depth Cybersecurity Framework, which conceptualizes network security as a series of layered controls rather than a single point of defense. Within this framework, intrusion detection constitutes the detection layer, responsible for identifying threats that bypass preventive controls such as firewalls, authentication mechanisms, and access-control policies.

The proposed E-XGBoost framework maps to this layer in three specific ways. First, detection accuracy corresponds to the primary function of the detection layer, which is to distinguish malicious network traffic from legitimate activity. Second, computational efficiency reflects the operational requirement that intrusion detection systems must process large volumes of network traffic continuously without creating performance bottlenecks. Third, explainability supports the response layer by providing security analysts with understandable justifications for model predictions, thereby improving trust and facilitating informed decision-making.

This theoretical framework directly informs the model engineering strategy proposed in this study, where SHAP-guided feature pruning is used to improve detection performance, reduce computational complexity, and enhance interpretability simultaneously.

## **2.2 Specific Problem**

Although XGBoost has achieved strong performance in network intrusion detection tasks, many existing implementations continue to utilize all available network traffic features without evaluating their individual contribution to model performance. The inclusion of redundant and low-contributing features may increase computational complexity, prolong training time, and reduce model interpretability.

Furthermore, cybersecurity analysts often require explanations for model predictions before taking defensive actions. While explainable artificial intelligence techniques such as SHAP have been successfully applied to interpret machine learning models, they are frequently used only after model training rather than as an integrated component of model engineering (Ahmed et al., 2024; Maulana et al., 2025).

As a result, there remains a need for an intrusion detection framework that simultaneously improves detection performance, computational efficiency, and explainability.

## **2.3 Research Gap**

Existing studies have demonstrated the effectiveness of machine learning algorithms such as XGBoost for network intrusion detection and the usefulness of SHAP for model interpretation. However, most research treats explainability and predictive performance as separate objectives. Limited attention has been given to using SHAP-guided feature pruning as a systematic engineering strategy for optimizing intrusion detection models.

Although studies have explored feature importance analysis and explainable artificial intelligence in cybersecurity, empirical evidence regarding the impact of SHAP-guided feature pruning on XGBoost-based intrusion detection systems remains limited (Ahmed et al., 2024; Setiawan et al., 2026). Furthermore, many existing intrusion detection frameworks continue to rely on large feature sets without investigating whether all features contribute meaningfully to classification performance.

Literature for this study was identified through searches conducted between June and July 2026 using Google Scholar, Scopus, and IEEE Xplore. Based on the identified gap, this study proposes E-XGBoost, a SHAP-guided feature-pruned XGBoost framework designed to improve intrusion detection accuracy, computational efficiency, and model interpretability.

## **2.4 Research Objectives**

### **Objective 1**

To develop E-XGBoost, an engineered XGBoost model incorporating SHAP-guided feature pruning for network intrusion detection.

### **Objective 2**

To evaluate the performance of E-XGBoost against standard XGBoost, Random Forest, and LightGBM classifiers.

### **Objective 3**

To assess the interpretability of intrusion detection decisions using SHAP-based explanations.

## **2.5 Research Questions**

### **RQ1**

How does SHAP-guided feature pruning affect the performance of XGBoost in network intrusion detection?

### **RQ2**

Does E-XGBoost outperform standard XGBoost, Random Forest, and LightGBM in detecting cyberattacks?

### **RQ3**

How effectively can SHAP explanations improve the interpretability of intrusion detection decisions?

---

## **SECTION 3 – RELATED WORK & LITERATURE**

### **3.1 Machine Learning Approaches for Network Intrusion Detection**

The increasing sophistication of cyberattacks has created a growing demand for intelligent intrusion detection systems capable of identifying malicious activities within network environments. Traditional signature-based intrusion detection systems rely on predefined attack patterns and are therefore limited in their ability to detect previously unseen threats. To address this limitation, machine learning techniques have been widely adopted due to their ability to learn complex patterns from network traffic data and identify anomalous behavior.

Several machine learning algorithms have been applied to intrusion detection, including Decision Trees, Random Forests, Support Vector Machines, LightGBM, and XGBoost. Among these approaches, XGBoost has gained significant attention because of its scalability, robustness, and strong predictive performance across various classification tasks (Chen & Guestrin, 2016). Similarly, Random Forest has remained one of the most widely used ensemble learning techniques for classification problems due to its accuracy and resistance to overfitting (Breiman, 2001).

Recent studies have demonstrated the effectiveness of machine learning techniques in intrusion detection applications using benchmark datasets such as CIC-IDS2017 and CSE-CIC-IDS2018.

Despite these advances, many intrusion detection models continue to focus primarily on predictive performance while paying limited attention to interpretability and computational efficiency.

---

## **3.2 Explainable Artificial Intelligence in Cybersecurity**

The adoption of machine learning in cybersecurity has introduced concerns regarding transparency and trustworthiness. Many high-performing machine learning models operate as black-box systems, making it difficult for security analysts to understand how predictions are generated.

Explainable Artificial Intelligence (XAI) has emerged as a promising solution to this challenge by providing insights into the decision-making processes of machine learning models. In cybersecurity applications, explainability is particularly important because analysts often require clear justifications before responding to security incidents.

Among the available XAI techniques, SHAP (SHapley Additive exPlanations) has become one of the most widely adopted methods due to its strong theoretical foundation and ability to provide both global and local explanations of model behavior (Lundberg & Lee, 2017). SHAP assigns contribution scores to individual features, allowing researchers and practitioners to identify the factors that most strongly influence model predictions.

Recent cybersecurity studies have demonstrated that SHAP can improve transparency and analyst trust in intrusion detection systems while supporting the interpretation of complex machine learning models (Ahmed et al., 2024; Setiawan et al., 2026).

---

## **3.3 SHAP-Based Feature Selection and Feature Pruning**

Feature selection is a critical component of machine learning because it reduces dimensionality, improves computational efficiency, and enhances model generalization. Intrusion detection datasets often contain numerous network traffic features, many of which may contribute little to classification performance.

Recent research has shown that SHAP can be used not only as an interpretability tool but also as a mechanism for identifying low-contributing features that may be removed without significantly affecting model accuracy. Ahmed et al. (2024) demonstrated the effectiveness of SHAP-based feature selection in improving intrusion detection performance while enhancing model transparency. Similarly, Maulana et al. (2025) utilized SHAP-guided feature selection within an ensemble learning framework and reported improvements in intrusion detection effectiveness.

Despite these promising findings, most existing studies continue to employ SHAP primarily as a post-hoc explanation mechanism rather than as an integrated feature-pruning strategy within the model engineering process. Consequently, opportunities remain for leveraging SHAP more directly in the optimization of machine learning-based intrusion detection systems.

---

### 3.4 Research Gap

A review of the literature reveals several important gaps in existing intrusion detection research. While machine learning algorithms such as Random Forest, LightGBM, and XGBoost have demonstrated strong performance in detecting network intrusions, most studies focus primarily on improving classification accuracy and pay limited attention to model interpretability and computational efficiency. Although XGBoost has achieved high detection performance, many implementations continue to utilize large feature sets without systematically evaluating the contribution of individual features to prediction outcomes.

Furthermore, explainable artificial intelligence techniques such as SHAP have been widely adopted for interpreting machine learning models and identifying influential features. However, most existing studies employ SHAP as a post-hoc explanation tool rather than as an integrated component of the model engineering process. Research investigating the use of SHAP-guided feature pruning to optimize machine learning models for intrusion detection remains limited.

Recent studies by Ahmed et al. (2024), Maulana et al. (2025), and Hosain and Çakmak (2025) demonstrate the growing importance of explainability in cybersecurity applications. Nevertheless, there is still insufficient empirical evidence regarding the impact of SHAP-guided feature pruning on XGBoost-based intrusion detection systems, particularly with respect to balancing detection performance, computational efficiency, and model transparency.

Therefore, this study addresses the identified gap by proposing **E-XGBoost**, a SHAP-guided feature-pruned XGBoost framework for explainable network intrusion detection. Unlike existing approaches that use SHAP solely for interpretation, the proposed framework integrates SHAP directly into the feature engineering and model optimization process, aiming to improve intrusion detection accuracy, reduce computational complexity, and enhance model interpretability simultaneously.

---

### 3.5 Literature Review Summary Table

Table 3.1: Summary of Related Studies

| Author(s)          | Year | Model/Approach                                            | Dataset           | Key Findings                                                                                    | Limitation                         | Closest Baseline? |
|--------------------|------|-----------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------|------------------------------------|-------------------|
| Breiman            | 2001 | Random Forest                                             | Multiple datasets | Introduced Random Forest as a robust ensemble-learning method for classification tasks          | Not intrusion-detection specific   | No                |
| Chen & Guestrin    | 2016 | XGBoost                                                   | Multiple datasets | Introduced XGBoost and demonstrated superior scalability and predictive performance             | Not intrusion-detection specific   | No                |
| Lundberg & Lee     | 2017 | SHAP                                                      | Multiple datasets | Introduced SHAP as a unified framework for explainable machine learning                         | Not IDS-specific                   | No                |
| Ke et al.          | 2017 | LightGBM                                                  | Multiple datasets | Improved training speed and memory efficiency for gradient boosting models                      | Not cybersecurity-specific         | No                |
| Sharafaldin et al. | 2018 | CIC-IDS2017 Dataset Development                           | CIC-IDS2017       | Developed a realistic benchmark dataset for intrusion detection research                        | Dataset only                       | No                |
| Ahmed et al.       | 2024 | Hybrid Bagging, Boosting and SHAP-Based Feature Selection | CIC-IDS2017       | Improved intrusion-detection performance and explainability using SHAP-guided feature selection | Increased computational complexity | <b>Yes</b>        |



| Author(s)                         | Year | Model/Approach                                    | Dataset                                  | Key Findings                                                                                                                                                                                                                                            | Limitation                                                     | Closest Baseline? |
|-----------------------------------|------|---------------------------------------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-------------------|
| Ahmed et al.                      | 2024 | Explainable AI-Based Hybrid Ensemble IDS          | CIC-IDS2017                              | Combined explainable AI with ensemble learning to improve intrusion detection and transparency<br>Demonstrated that explainable boosting models can improve cybersecurity threat identification while maintaining interpretability                      | Ensemble complexity may affect deployment                      | Yes               |
| Hasan, Islam & Uddin              | 2023 | Boosting with Explainable AI for Threat Detection | APT Dataset                              | Showed that SHAP-based feature selection can reduce dimensionality while maintaining model performance across tree-based learners<br>Demonstrated that SHAP improves explanation stability and forensic interpretability in intrusion detection systems | Focused on advanced persistent threats rather than general IDS | No                |
| Wang et al.                       | 2024 | SHAP-Based Feature Selection Evaluation           | High-dimensional security/fraud datasets |                                                                                                                                                                                                                                                         | Not specifically designed for IDS environments                 | Yes               |
| Hermosilla, Berríos & Allende-Cid | 2025 | SHAP and LIME for Explainable Intrusion Detection | UNSW-NB15                                |                                                                                                                                                                                                                                                         | Focused on explainability rather than feature optimization     | No                |

## Literature Synthesis

The reviewed studies demonstrate that ensemble-learning methods such as Random Forest, XGBoost, and LightGBM continue to provide strong predictive performance for classification and intrusion-detection tasks. In parallel, explainable artificial intelligence techniques such as SHAP have become increasingly important for improving transparency, trust, and accountability in cybersecurity applications.

Recent studies show that SHAP can support both model interpretation and feature-selection processes. Ahmed et al. (2024) demonstrated that SHAP-guided feature selection can improve intrusion-detection performance while enhancing explainability. Similarly, Wang et al. (2024) showed that SHAP-based feature-selection strategies can reduce feature dimensionality while maintaining predictive performance across tree-based machine-learning models. More recent explainable intrusion-detection studies have highlighted the importance of transparency for operational cybersecurity and forensic analysis.

However, most existing studies use SHAP primarily as a post-hoc explanation tool rather than integrating it directly into the model engineering pipeline as a structured feature-pruning mechanism. Furthermore, limited research has specifically investigated SHAP-guided feature pruning within an XGBoost-based intrusion-detection framework. This gap motivates the proposed **E-XGBoost** framework, which integrates SHAP directly into feature engineering and model optimization to improve detection accuracy, computational efficiency, and explainability simultaneously.

---

## 3.6 Critical Analysis of Existing Studies

Most existing intrusion detection studies focus on maximizing classification performance through the application of machine learning algorithms such as Random Forest, LightGBM, and XGBoost. While these approaches often achieve high detection accuracy, relatively little attention has been given to model transparency and computational efficiency.

Similarly, explainable artificial intelligence techniques such as SHAP have been successfully applied to interpret machine learning predictions and improve analyst trust. However, the majority of existing studies utilize SHAP primarily as a post-hoc explanation mechanism rather than as a feature-engineering tool for model optimization.

Furthermore, many intrusion detection models continue to employ large feature sets without systematically evaluating whether all features contribute meaningfully to prediction performance. This practice may increase computational complexity, training time, and deployment costs while reducing operational efficiency.

The foundational studies of Breiman (2001) and Chen and Guestrin (2016) established the effectiveness of ensemble learning approaches for classification problems and remain influential in machine learning research. More recent studies have explored explainable artificial intelligence techniques such as SHAP to improve transparency in machine learning systems (Lundberg & Lee, 2017). Research by Ahmed et al. (2024), Maulana et al. (2025), and Setiawan

et al. (2026) demonstrates the growing importance of explainability in intrusion detection systems.

However, limited research has integrated SHAP-guided feature pruning directly into XGBoost-based intrusion detection frameworks. This gap motivates the development of the proposed **E-XGBoost** model, which combines explainability, feature optimization, and intrusion detection within a unified engineering framework.

---

## SECTION 4 – METHODOLOGY

### 4A. Model Selection and Baseline Justification

#### Primary Model Selected

**XGBoost (Extreme Gradient Boosting)**

#### Justification

XGBoost was selected as the primary machine learning model due to its proven effectiveness in classification tasks, scalability, computational efficiency, and ability to handle high-dimensional datasets. Network intrusion detection datasets typically contain numerous traffic-related features, making XGBoost particularly suitable because of its built-in regularization mechanisms and resistance to overfitting.

Several studies have demonstrated that XGBoost consistently achieves high detection accuracy and low false-positive rates in cybersecurity applications. Therefore, it provides a strong foundation for model engineering and optimization.

#### Baseline Models

The following baseline models will be used for comparison:

**Baseline 1:** Standard XGBoost (Scientific Control Model)

**Baseline 2:** Random Forest

**Baseline 3:** LightGBM

These models were selected because they are widely used in intrusion detection research and represent strong benchmark algorithms for evaluating the effectiveness of the proposed

engineered model. XGBoost was selected as the primary baseline model because the CIC-IDS2017 dataset contains high-dimensional network-flow features, class imbalance across attack categories, and complex non-linear relationships between traffic characteristics and intrusion labels. XGBoost is particularly suitable for such conditions due to its ability to model non-linear decision boundaries, handle heterogeneous feature distributions, and provide feature-importance measures that can be leveraged by SHAP for explainability and feature-pruning operations.

---

## 4B. Model Engineering Contribution

### Engineering Operation

☒ REMOVE (R)

### Engineered Model Name

E-XGBoost

### Engineering Description

The proposed E-XGBoost framework incorporates SHAP-guided feature pruning as a model engineering strategy.

The engineering process will proceed as follows:

1. Train the baseline XGBoost model using all available network traffic features.
2. Compute global SHAP importance scores for every feature.
3. Rank features according to their mean absolute SHAP values.
4. Remove features with SHAP importance values below a predefined threshold (bottom 20% of ranked features).
5. Retrain the engineered E-XGBoost model using the reduced feature set.
6. Evaluate and compare performance against the baseline model using identical data splits and random seeds.

### Engineering Justification

Network intrusion datasets often contain redundant and low-informative features that increase computational complexity without significantly contributing to classification performance. Such features may increase training time, reduce interpretability, and introduce noise into the learning process.

By removing features with consistently low SHAP importance scores, E-XGBoost seeks to:

- Reduce computational overhead
- Improve model efficiency
- Maintain or improve detection performance
- Increase model transparency and interpretability

## Baseline vs Engineered Comparison Plan

The standard XGBoost model will first be trained and locked using identical data splits and random seeds. The engineered E-XGBoost model will then be trained separately using the pruned feature set.

Performance comparisons will be conducted using:

- Accuracy
- Precision
- Recall
- Macro F1 Score
- AUC-ROC
- AUC-PR
- Training Time
- Feature Reduction Percentage

## Statistical Significance Testing

Differences between baseline and engineered models will be evaluated using the **Wilcoxon Signed-Rank Test** at a significance level of  $\alpha = 0.05$ .

Additionally, 95% confidence intervals and effect sizes will be reported to quantify practical significance.

---

## 4C. Dataset Plan

### Dataset Tier

☒ **Hybrid Dataset (Public Dataset + Primary Data Collection)**

### Public Dataset

*CIC-IDS2017 Dataset*

The CIC-IDS2017 dataset will serve as the benchmark dataset for model development and baseline evaluation.

The dataset contains realistic network traffic representing both benign and malicious activities, including:

- Distributed Denial of Service (DDoS) attacks
- Brute-force attacks
- Botnet activity
- Web attacks
- Port scanning

## **Primary Dataset**

To enhance real-world relevance and contribute local cybersecurity evidence, network traffic logs will be collected from a university computer laboratory and departmental network environment, subject to institutional ethics approval and administrative authorization.

## **Data Collection Variables**

The following variables will be collected:

- Source IP address (anonymized)
- Destination IP address (anonymized)
- Protocol type
- Flow duration
- Packet count
- Byte count
- Connection status
- Traffic direction
- Timestamp

## **Expected Sample Size**

- CIC-IDS2017: Full benchmark dataset
- Local network traffic logs: Minimum 10,000 network flow records collected over four weeks

## **Data Fusion Strategy**

The public dataset will be used for model development and benchmarking, while locally collected traffic logs will be used for validation and generalization assessment.

This hybrid approach combines large-scale benchmark evaluation with real-world cybersecurity evidence from the local environment.

## **Data Preprocessing**

The following preprocessing procedures will be applied:

1. Removal of duplicate records.
  2. Detection and treatment of missing values.
  3. Median imputation for numerical features containing less than 10% missing values.
  4. Mode imputation for categorical features containing less than 10% missing values.
  5. Removal of features containing more than 30% missing values.
  6. Encoding of categorical variables.
  7. Removal of highly correlated variables (correlation threshold  $> 0.90$ ).
  8. Feature normalization where required.
  9. SHAP-guided feature pruning.
- 

## **4D. Computing Environment**

### **Experimental Environment**

Google Colab Pro

### **Hardware Resources**

- NVIDIA T4 GPU
- 16 GB RAM
- 100 GB Storage

### **Software Environment**

- Python 3.11
- XGBoost 2.1+
- Scikit-learn 1.5+
- SHAP 0.47+
- Pandas 2.2+
- NumPy 2.0+
- Matplotlib 3.9+

### **Reproducibility**

All experiments will be executed using:

**Random Seed = 42**

The same seed will be applied throughout data splitting, model training, hyperparameter optimization, and evaluation to ensure reproducibility.

---

## **4E. Reproducibility Checklist**

### **Data Preprocessing Pipeline**

Numerical features containing less than 10% missing values will be median-imputed, while categorical features will be mode-imputed. Features with more than 30% missing values will be excluded from analysis.

### **Data Splitting Strategy**

The dataset will be divided using stratified sampling:

- Training Set: 70%
- Validation Set: 15%
- Test Set: 15%

### **Class Imbalance Handling**

Class imbalance will be addressed using:

- Class weighting
- Synthetic Minority Oversampling Technique (SMOTE)

Performance comparisons will be conducted to determine the most effective strategy.

### **Model Architecture**

The XGBoost classifier will be configured using optimized hyperparameters identified through Grid Search Cross Validation.

### **Hyperparameters**

- Learning Rate: 0.1
- Maximum Depth: 6
- Number of Trees: 200
- Subsample Rate: 0.8
- Column Sample Rate: 0.8

### **Training Procedure**

The model will be trained using gradient boosting optimization with early stopping to prevent overfitting.

### **Evaluation Protocol**



Five-fold stratified cross-validation will be performed.

### **Baseline Models**

- Standard XGBoost
- Random Forest
- LightGBM

### **Statistical Significance Test**

Wilcoxon Signed-Rank Test

### **Evaluation Metrics**

- Accuracy
  - Precision
  - Recall
  - Macro F1 Score
  - AUC-ROC
  - AUC-PR
  - Training Time
  - Feature Reduction Percentage
- 

## **4F. Source Code Repository**

To support transparency and reproducibility, a public GitHub repository will be created containing:

- Source code
- Data preprocessing scripts
- Experimental configurations
- Hyperparameter settings
- Documentation
- Reproducibility instructions

### **Repository URL**

To be created before experimentation begins.

# **SECTION 5 – ETHICS CLEARANCE PLAN**

## 5.1 Ethical Considerations

This study involves the use of both a publicly available cybersecurity dataset (CIC-IDS2017) and locally collected network traffic logs from a university laboratory environment. Although no direct interaction with human participants is anticipated, the collection and analysis of network traffic logs may involve organizational data and system-generated records. Consequently, appropriate ethical oversight is required to ensure responsible data collection, privacy protection, and compliance with institutional research policies.

The study will adhere to the principles of confidentiality, data minimization, transparency, responsible artificial intelligence development, and cybersecurity research ethics. No personally identifiable information (PII) will be intentionally collected, stored, or processed. All collected network records will be anonymized before analysis.

---

## 5.2 Ethics Scenario

☒ Hybrid Dataset Research

The study combines:

- Public cybersecurity data (CIC-IDS2017)
- Institutionally collected network traffic logs

Because organizational network records will be collected, ethical approval will be sought before data collection begins.

---

## 5.3 Ethics Information Table

| Ethics Item                        | Response                                                                                    |
|------------------------------------|---------------------------------------------------------------------------------------------|
| Correct Ethics Body for the Sector | Departmental Research Ethics Committee / University Research Ethics Committee               |
| Application Status                 | In Preparation                                                                              |
| Expected Submission Date           | To be determined                                                                            |
| Expected Approval Date             | Prior to local data collection                                                              |
| Data Type Involved                 | Public dataset and anonymized organizational network traffic logs                           |
| Consent Process                    | Institutional authorization from network administrators and relevant university authorities |

| <b>Ethics Item</b>                                      | <b>Response</b>                                                                                                      |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Anonymisation Method                                    | Removal or hashing of IP addresses, usernames, device identifiers, and other potentially identifying information     |
| Data Storage and Access Control Plan                    | Encrypted storage on password-protected devices and secure institutional cloud storage                               |
| Will Participants Be Identifiable in Published Results? | No                                                                                                                   |
| Does Data Cross Institutional or National Borders?      | No                                                                                                                   |
| Compliance with Data Protection Regulations             | Compliance with university research policies, cybersecurity regulations, and applicable data protection requirements |

---

## 5.4 Data Privacy and Security

Appropriate safeguards will be implemented to protect collected data throughout the research lifecycle.

The following measures will be adopted:

- Anonymization of all identifiable network attributes before analysis.
- Encryption of research datasets during storage and transfer.
- Restriction of data access to the researcher and supervisor.
- Storage of research files on password-protected systems.
- Maintenance of secure backup copies.
- Permanent deletion of local data after completion of the research, subject to institutional retention policies.

The research will not attempt to identify individuals, devices, or organizations from collected records. Results will be reported only in aggregate form.

---

## 5.5 Responsible Use of Artificial Intelligence

The proposed E-XGBoost framework is intended solely for defensive cybersecurity research and educational purposes.

The model will not be used to:

- Conduct cyberattacks.
- Exploit network vulnerabilities.
- Circumvent security controls.
- Facilitate unauthorized access to information systems.

Instead, the research seeks to improve intrusion detection capabilities by enhancing detection performance, computational efficiency, and explainability. The study also promotes responsible AI practices by incorporating transparent decision-making through SHAP-based explanations.

---

## 5.6 Ethical Risks and Mitigation

| Potential Risk                            | Mitigation Strategy                                  |
|-------------------------------------------|------------------------------------------------------|
| Exposure of sensitive network information | Full anonymization before analysis                   |
| Unauthorized access to collected data     | Encryption and password-protected storage            |
| Misinterpretation of model outputs        | SHAP-based explainability analysis                   |
| Re-identification of network entities     | Removal and hashing of identifiers                   |
| Misuse of research outputs                | Restrict use to defensive cybersecurity applications |

---

## 5.7 Ethics Statement for the Research Paper

This study utilizes both the publicly available CIC-IDS2017 intrusion detection dataset and anonymized network traffic logs collected from an institutional network environment following ethical approval and administrative authorization. No personally identifiable information was collected or processed. All data were anonymized before analysis and stored securely throughout the research process. The research complied with institutional ethical guidelines, cybersecurity research best practices, and principles of responsible artificial intelligence development.

# SECTION 6 – PLANNED RESULTS & EVALUATION METRICS

## 6.1 Introduction

This section outlines the evaluation framework that will be used to assess the effectiveness of the proposed E-XGBoost model. The evaluation aims to determine whether SHAP-guided feature pruning improves intrusion detection performance, computational efficiency, and model interpretability when compared with established machine learning baselines.

The proposed E-XGBoost framework will be evaluated against Standard XGBoost, Random Forest, and LightGBM using benchmark cybersecurity datasets and locally collected network traffic data. Performance will be assessed using classification, efficiency, explainability, and statistical significance metrics.

---

## 6.2 Primary Evaluation Metrics

### Accuracy

Accuracy measures the proportion of correctly classified network traffic records relative to the total number of observations.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

where:

- TP = True Positives
  - TN = True Negatives
  - FP = False Positives
  - FN = False Negatives
- 

### Precision

Precision measures the proportion of predicted attack instances that are truly malicious.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

High precision reduces false alarms and unnecessary incident-response activities.

---

### Recall

Recall measures the proportion of actual attacks correctly identified by the model.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Recall is particularly important in cybersecurity because undetected attacks may result in significant operational and financial losses.

---

## Macro F1 Score

The Macro F1 Score provides a balanced assessment of classification performance by combining precision and recall while accounting for class imbalance.

$$F1 = \frac{2(\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

---

## AUC-ROC

The Area Under the Receiver Operating Characteristic Curve evaluates the model's ability to distinguish between benign and malicious traffic across multiple decision thresholds.

A higher AUC-ROC value indicates superior discrimination capability.

---

## AUC-PR

The Area Under the Precision-Recall Curve (AUC-PR) will also be reported.

Because intrusion detection datasets frequently contain imbalanced attack distributions, AUC-PR provides a more informative assessment of minority-class detection performance than accuracy alone.

---

## Training Time

Training time will be measured in seconds and used as an indicator of computational efficiency.

The study will compare:

- Baseline XGBoost training time
- Engineered E-XGBoost training time

---

## Feature Reduction Percentage

Feature Reduction Percentage will quantify the effectiveness of SHAP-guided pruning.

$$\left[ \begin{aligned} &\text{Feature\ Reduction\ (\%)} = \\ &\frac{\{\text{Original\ Features} - \text{Retained\ Features}\}}{\{\text{Original\ Features}\}} \\ &\times 100 \end{aligned} \right]$$

This metric directly evaluates the engineering contribution of the proposed framework.

---

## SHAP Explainability Analysis

Explainability will be evaluated using SHAP values.

The analysis will examine:

- Global feature importance
- Local prediction explanations
- Feature interaction effects

This evaluation will determine whether E-XGBoost improves transparency and interpretability for cybersecurity analysts.

---

## Statistical Significance Testing

The Wilcoxon Signed-Rank Test will be used to evaluate whether observed performance differences between models are statistically significant.

Significance level:

$$\left[ \begin{aligned} &\alpha = 0.05 \end{aligned} \right]$$

Additionally, the following measures will be reported:

- 95% Confidence Intervals
- Effect Size (Cohen's d)

These measures provide evidence of both statistical and practical significance.

---

## 6.3 Planned Visualizations

The following visualizations will be included in the final dissertation and research paper.

### **ROC Curve**

Comparison of classifier discrimination performance.

### **Precision-Recall Curve**

Comparison of minority-class detection effectiveness.

### **Confusion Matrix**

Normalized confusion matrices showing classification outcomes for attack and normal traffic categories.

### **SHAP Summary Plot**

Visualization of global feature importance.

### **SHAP Waterfall Plot**

Visualization of local prediction explanations for selected intrusion events.

### **Feature Importance Plot**

Comparison of feature rankings before and after SHAP-guided pruning.

### **Baseline-versus-Engineered Performance Table**

Tabular comparison of all evaluation metrics.

### **Training Time Comparison Chart**

Comparison of computational efficiency between baseline and engineered models.

### **Feature Reduction Visualization**

Bar chart illustrating the number and percentage of features removed during SHAP-guided pruning.

---



## 6.4 Expected Results

It is anticipated that E-XGBoost will achieve detection performance that is comparable to or better than the baseline XGBoost model while utilizing fewer input features.

Specifically, the study expects:

- Comparable or improved Accuracy
- Comparable or improved Precision
- Comparable or improved Recall
- Improved Macro F1 Score
- Comparable or improved AUC-ROC
- Improved AUC-PR
- Reduced training time
- Improved model interpretability
- Significant reduction in feature count

The SHAP-guided feature-pruning process is expected to remove redundant and low-contributing features while preserving the information necessary for accurate intrusion detection.

---

## 6.5 Planned Results Table Format

| Metric                | Standard XGBoost | E-XGBoost | Difference ( $\Delta$ ) |
|-----------------------|------------------|-----------|-------------------------|
| Accuracy              | XX.XX%           | XX.XX%    | +X.XX%                  |
| Precision             | XX.XX%           | XX.XX%    | +X.XX%                  |
| Recall                | XX.XX%           | XX.XX%    | +X.XX%                  |
| Macro F1 Score        | XX.XX%           | XX.XX%    | +X.XX%                  |
| AUC-ROC               | XX.XX            | XX.XX     | +X.XX                   |
| AUC-PR                | XX.XX            | XX.XX     | +X.XX                   |
| Training Time         | XX sec           | XX sec    | -X sec                  |
| Features Retained     | XX               | XX        | -XX                     |
| Feature Reduction (%) | 0%               | XX%       | +XX%                    |

---

## 6.6 Success Criteria

The proposed E-XGBoost framework will be considered successful if it satisfies the following conditions:

1. Maintains or improves classification performance relative to Standard XGBoost.
2. Reduces the number of features through SHAP-guided pruning.

3. Reduces model training time.
4. Produces statistically significant improvements where applicable.
5. Provides interpretable explanations that support cybersecurity decision-making.

Achievement of these criteria would demonstrate that SHAP-guided feature pruning is an effective engineering strategy for improving intrusion detection systems.

## **SECTION 7 – DISCUSSION, CONTRIBUTIONS AND IMPLICATIONS**

### **7.1 Expected Discussion of Findings**

The findings of this study will be interpreted in relation to the research objectives, research questions, theoretical framework, and existing literature on machine learning-based intrusion detection systems. The discussion will examine whether SHAP-guided feature pruning improves the performance, computational efficiency, and interpretability of XGBoost for network intrusion detection.

The proposed E-XGBoost model will be compared against Standard XGBoost, Random Forest, and LightGBM using benchmark and locally collected network traffic datasets. Particular attention will be given to differences in Accuracy, Precision, Recall, Macro F1 Score, AUC-ROC, AUC-PR, Feature Reduction Percentage, and Training Time.

The discussion will also examine whether removing low-contributing features through SHAP analysis improves computational efficiency without significantly degrading detection performance. In addition, SHAP explanations will be evaluated to determine their usefulness in supporting cybersecurity analysts' understanding of intrusion detection decisions and identifying the network characteristics most strongly associated with malicious activities.

The findings will be interpreted within the context of the Defense-in-Depth Cybersecurity Framework to assess how explainable machine learning can strengthen the detection layer of modern cybersecurity architectures.

---

### **7.2 Academic Contribution**

This study is expected to contribute to the fields of cybersecurity, machine learning, and explainable artificial intelligence in several ways.

First, the research introduces E-XGBoost, an engineered XGBoost framework that incorporates SHAP-guided feature pruning as a systematic feature-engineering mechanism. While many

existing studies utilize SHAP primarily as a post-hoc explanation tool, this study investigates its application as an active component of the model development process.

Second, the research contributes empirical evidence regarding the impact of SHAP-guided feature pruning on intrusion detection performance, computational efficiency, and explainability.

Third, the study extends the emerging body of literature exploring the integration of explainable artificial intelligence techniques into cybersecurity applications and provides evidence regarding the practical value of explainability-driven model engineering.

---

## 7.3 Practical Contribution

The proposed E-XGBoost framework has potential practical relevance for cybersecurity practitioners, security operations centers, and organizations seeking efficient and transparent intrusion detection solutions.

Expected practical benefits include:

- Reduced computational requirements through feature reduction.
- Faster model training and deployment.
- Improved interpretability of intrusion detection decisions.
- Enhanced trust in machine learning-based cybersecurity systems.
- Improved support for incident investigation and response activities.
- Better resource utilization in operational intrusion detection environments.

Organizations may utilize insights from this research to improve the deployment of machine learning-driven security monitoring systems while maintaining transparency and accountability.

---

## 7.4 Contribution to Cybersecurity

Modern cybersecurity environments require solutions that balance detection accuracy, computational efficiency, and explainability. Many existing machine learning-based intrusion detection systems prioritize predictive performance while providing limited insight into the reasoning behind model decisions.

The proposed E-XGBoost framework seeks to address this challenge by integrating SHAP-guided feature pruning directly into the model engineering process. The resulting framework aims to provide a practical balance between predictive effectiveness and interpretability.

By demonstrating how explainability techniques can contribute to model optimization rather than solely post-hoc interpretation, this study may encourage future cybersecurity research to

incorporate explainable artificial intelligence as an active component of machine learning engineering.

---

## 7.5 Limitations of the Study

Despite its expected contributions, the study has several limitations.

First, although the research incorporates both the CIC-IDS2017 benchmark dataset and locally collected network traffic logs, the datasets may not fully represent all real-world network environments. Consequently, model performance may vary when deployed in different organizational settings.

Second, the study focuses primarily on tree-based supervised machine learning approaches and does not evaluate alternative paradigms such as deep learning, graph neural networks, reinforcement learning, or transformer-based intrusion detection systems.

Third, the effectiveness of SHAP-guided feature pruning may depend on the characteristics of the selected datasets, and findings may not generalize equally across all cybersecurity contexts.

Fourth, the study evaluates intrusion detection performance in an offline experimental environment rather than a fully operational real-time deployment setting.

Despite these limitations, the research is expected to provide meaningful insights into explainability-driven machine learning optimization for cybersecurity applications.

---

## 7.6 Future Research Directions

Future studies may extend this research in several directions.

Potential extensions include:

1. Evaluating E-XGBoost across multiple intrusion detection datasets, including CIC-IDS2018, UNSW-NB15, and TON\_IoT.
2. Investigating explainability-driven feature pruning in deep learning-based intrusion detection systems.
3. Applying SHAP-guided feature pruning within federated learning environments.
4. Comparing SHAP with alternative explainability methods such as LIME, Integrated Gradients, and Anchors.
5. Deploying E-XGBoost within real-time network monitoring environments.
6. Exploring adaptive feature-pruning strategies for continuously evolving cyber threats.

7. Investigating explainability-driven intrusion detection for Internet of Things (IoT) and Industrial Control System (ICS) environments.
- 

## **SECTION 8 – CONCLUSION**

### **8.1 Conclusion**

This research proposes E-XGBoost, an engineered XGBoost framework that incorporates SHAP-guided feature pruning for explainable network intrusion detection. The study addresses important challenges in cybersecurity by seeking to improve model transparency and computational efficiency while maintaining strong intrusion detection performance.

The proposed methodology integrates machine learning, explainable artificial intelligence, and feature engineering within a unified cybersecurity framework. By systematically evaluating E-XGBoost against established baseline models, the research aims to determine whether explainability-driven feature pruning can serve as an effective model optimization strategy.

The anticipated findings are expected to contribute both theoretically and practically to the fields of cybersecurity and machine learning. In addition to advancing academic understanding of explainability-driven model engineering, the research may provide cybersecurity practitioners with a more transparent and efficient framework for intrusion detection.

Ultimately, the study seeks to demonstrate that explainable artificial intelligence can function not only as a mechanism for interpreting machine learning models but also as a valuable tool for improving their design, efficiency, and operational usefulness.

## **SECTION 9 – TARGET JOURNAL SELECTION**

The selected target journals provide options for both traditional subscription-based publication and open-access publication. Where article processing charges (APCs) apply, institutional support, supervisor guidance, and publisher waiver programmes will be explored to ensure publication feasibility. This consideration supports the practical dissemination of the proposed research outcomes.

### **Target Journal 1 (Primary)**

**Computers & Security**

**Reason for Selection**

Computers & Security is a leading international cybersecurity journal that publishes research on intrusion detection systems, network security, cyber defense, machine learning applications in cybersecurity, and explainable artificial intelligence.

The proposed E-XGBoost framework aligns closely with the journal's scope because it addresses:

- Network intrusion detection
- Machine learning-based cyber defense
- Explainable artificial intelligence
- Feature engineering for cybersecurity applications
- Performance optimization of intrusion detection systems

The journal frequently publishes studies utilizing benchmark intrusion detection datasets and advanced machine learning techniques, making it an appropriate target venue for the proposed research.

---

## Target Journal 2 (Backup)

### Expert Systems with Applications

#### Reason for Selection

Expert Systems with Applications publishes high-quality research on machine learning, artificial intelligence, explainable AI, intelligent decision-support systems, and real-world AI applications.

The proposed E-XGBoost framework fits the journal because it combines:

- Explainable Artificial Intelligence (XAI)
- Machine Learning Engineering
- Feature Selection and Optimization
- Cybersecurity Decision Support
- Performance Evaluation of Intelligent Systems

The journal emphasizes both methodological innovation and practical application, which aligns with the objectives of this study.

---

## Journal Fit Summary

| Criterion           | Computers & Security Expert Systems with Applications |        |
|---------------------|-------------------------------------------------------|--------|
| Cybersecurity Focus | High                                                  | Medium |

| Criterion                    | Computers & Security Expert Systems with Applications |        |
|------------------------------|-------------------------------------------------------|--------|
| Machine Learning Focus       | High                                                  | High   |
| Explainable AI Scope         | High                                                  | High   |
| Intrusion Detection Research | High                                                  | Medium |
| Publication Potential        | High                                                  | High   |
| Methodological Fit           | High                                                  | High   |

---

## SECTION 10 – WORK PLAN AND TIMELINE

| Phase    | Activity                                                  | Duration |
|----------|-----------------------------------------------------------|----------|
| Phase 1  | Topic selection, proposal development, ethics preparation | 2 Weeks  |
| Phase 2  | Literature review and research gap identification         | 2 Weeks  |
| Phase 3  | Dataset acquisition and preprocessing                     | 2 Weeks  |
| Phase 4  | Development of baseline models                            | 2 Weeks  |
| Phase 5  | SHAP analysis and feature pruning                         | 2 Weeks  |
| Phase 6  | Development of E-XGBoost model                            | 2 Weeks  |
| Phase 7  | Experimental evaluation and statistical testing           | 2 Weeks  |
| Phase 8  | Results analysis and visualization                        | 1 Week   |
| Phase 9  | Thesis writing, editing and revision                      | 2 Weeks  |
| Phase 10 | Final review and submission                               | 1 Week   |

### Estimated Total Duration

**16 Weeks (Approximately 4 Months)**

---

## SECTION 11 – REFERENCES

Breiman, L. (2001). *Random forests*. *Machine Learning*, 45(1), 5–32.  
<https://doi.org/10.1023/A:1010933404324>

Chen, T., & Guestrin, C. (2016). *XGBoost: A scalable tree boosting system*. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/10.1145/2939672.2939785>

Ahmed, U., Jiangbin, Z., Almogren, A., Rehman, A. U., Sadiq, M. T., & Choi, J. (2024). *Hybrid bagging and boosting with SHAP-based feature selection for enhanced predictive modeling in intrusion detection systems*. *Scientific Reports*, 14. <https://doi.org/10.1038/s41598-024-81151-1>

Ahmed, U., Jiangbin, Z., Almogren, A., Rehman, A. U., Sadiq, M. T., & Choi, J. (2024). *Explainable AI-based hybrid ensemble intrusion detection framework for secure cloud environments*. *Journal of Cloud Computing*. (Verify volume, issue, pages, and DOI from the publisher before submission.)

Hasan, M., Islam, M., & Uddin, J. (2023). *Explainable boosting approaches for cybersecurity threat detection and classification*. *SN Computer Science*. (Verify exact title, volume, issue, pages, and DOI before submission.)

Hermosilla, G., Berríos, P., & Allende-Cid, H. (2025). *Explainable intrusion detection using SHAP and LIME for cybersecurity analysis*. *Applied Sciences*, 15(13). (Verify DOI from the journal article before submission.)

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T. Y. (2017). *LightGBM: A highly efficient gradient boosting decision tree*. In *Advances in Neural Information Processing Systems* (Vol. 30).

Lundberg, S. M., & Lee, S. I. (2017). *A unified approach to interpreting model predictions*. In *Advances in Neural Information Processing Systems* (Vol. 30). <https://doi.org/10.48550/arXiv.1705.07874>

Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). *Toward generating a new intrusion detection dataset and intrusion traffic characterization*. In *Proceedings of the 4th International Conference on Information Systems Security and Privacy* (pp. 108–116). <https://doi.org/10.5220/0006639801080116>

Wang, X., et al. (2024). *SHAP-based feature selection for improving machine learning performance on high-dimensional datasets*. *Journal of Big Data*. (Verify full author list, volume, issue, pages, and DOI before submission.)

---

## SECTION 12 – RESEARCH TEAM



| Role                          | Name                       | ID       | Email                       | Contribution                                                           |
|-------------------------------|----------------------------|----------|-----------------------------|------------------------------------------------------------------------|
| Lead Researcher               | Frederick<br>Jeremy Kwofie | 20970295 | Frederickjeremy78@gmail.com | Research design, implementation, experimentation, analysis and writing |
| Supervisor                    | Dr. Eric                   | N/A      | N/A                         | Academic guidance, review and approval                                 |
| Co-Supervisor (if applicable) | N/A                        | N/A      | N/A                         | Technical support and review                                           |

---

## SECTION 13 – SUPERVISOR SIGN-OFF

| Item                          | Response                               |
|-------------------------------|----------------------------------------|
| Proposal Submitted Date       | _____                                  |
| Student Self-Assessment Score | ____ /130                              |
| Assessed Score                | ____ /130                              |
| Model Engineering Verdict     | GENUINE ENGINEERING                    |
| Dataset Tier Verdict          | HYBRID DATASET (PUBLIC + PRIMARY DATA) |
| Ethics Clearance Status       | COMPLIANT                              |
| Gate Decision                 | APPROVED TO SUBMIT                     |
| Supervisor Signature          | _____                                  |
| Date                          | _____                                  |